CE 310

Week 14 — Bootstrap Confidence Intervals

Quantifying Uncertainty Without Distributional Assumptions

University of Arizona · Dr. Wooyoung Jung · Fall 2026

Bootstrap vs. Monte Carlo

Both methods use repeated sampling to quantify uncertainty — but from different sources.

Week 13 — Monte Carlo

- Samples from *assumed* distributions
- Normal(0, 2°F), Normal(0.12, 0.01)
- Asks: if inputs vary like this, what does the output distribution look like?
- Requires: a model and distributional assumptions

Week 14 — Bootstrap

- Resamples from *your observed data*
- No distributional assumption
- Asks: how uncertain is the statistic I computed?
- Requires: only the data itself

Bootstrap answers a different question: not "what if?" but "how reliable is this estimate?"

The Bootstrap Algorithm

Three steps:

1. Start with your n observations — the original sample
2. Draw n observations **with replacement** → a bootstrap resample (same n, but some rows appear 2–3×, others not at all)
3. Compute the statistic of interest on this resample → record one bootstrap estimate

Repeat steps 2–3 N_BOOT = 1,000 times.

The distribution of 1,000 bootstrap estimates is the **bootstrap sampling distribution**. The 2.5th and 97.5th percentiles form the **95% bootstrap CI**.

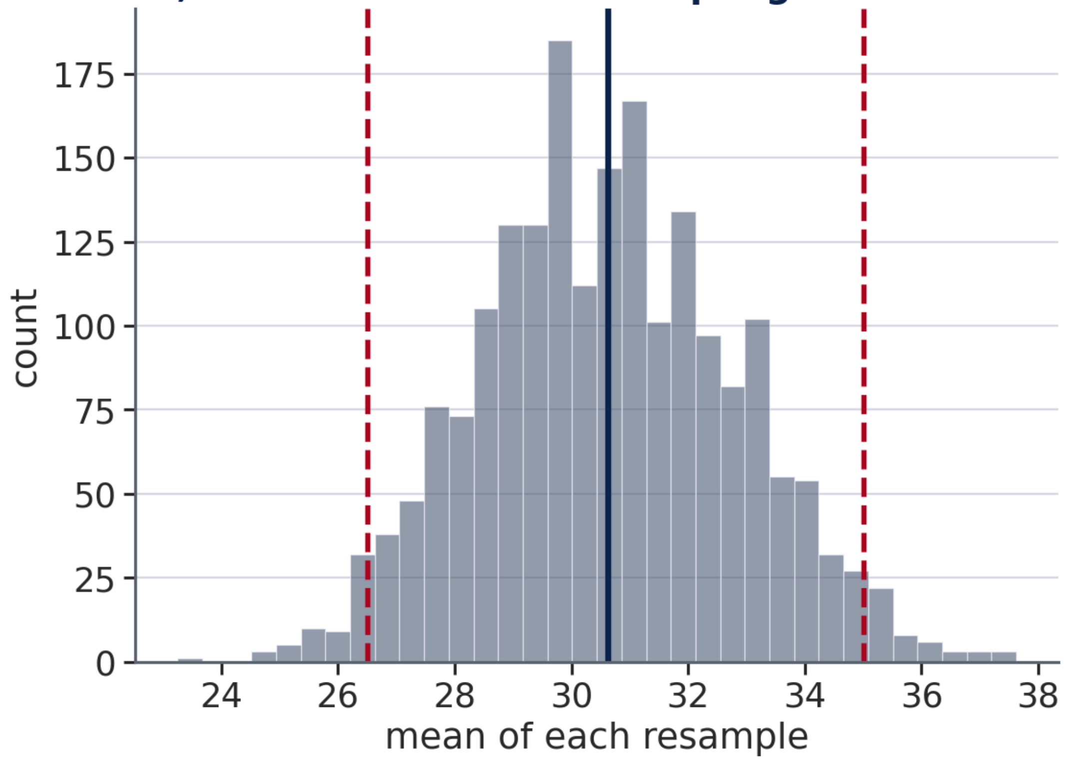
Drawing n items with replacement from n items picks up only about **63 %** of them — $1 - 1/e$. Every resample leaves roughly a third of your data on the table, and that is exactly where the variation comes from.

The Mechanism, on Eight Rows

Draw n rows with replacement — gold = a repeat

original sample	21	34	27	41	30	25	38	29	**mean 30.6**
resample 1	34	30	21	29	38	25	27	25	**mean 28.6**
resample 2	21	27	29	34	41	41	25	25	**mean 30.4**
resample 3	21	41	38	21	27	27	30	41	**mean 30.8**

2,000 of them make a sampling distribution



Each resample is the same size as the original but not the same contents — some rows arrive twice, others not at all. That reshuffling is the entire source of the variation.

Bootstrap CI for a Mean — Validation

For the mean, there is an analytical formula (the t-interval). Running bootstrap on the same statistic lets us check that both methods agree — validating the bootstrap before applying it to harder cases.

For mean outdoor temperature ($n = 8{,}760$):

- Observed mean: the sample mean of all 8,760 temperature readings
- Bootstrap std ≈ std_temp / sqrt(n) — the standard error of the mean
- Bootstrap 95% CI ≈ mean ± 2 × SE — a very tight interval for such large n
- Analytical t-interval: nearly identical

When bootstrap and t-interval agree closely, the normality assumption that the t-interval requires is also being met. Agreement here is expected and reassuring.

Quick Check — Bootstrap vs. Analytical CI

60 seconds · predict the agreement

Bootstrap resamples the data 1,000 times and computes the mean outdoor temperature each time, building an empirical distribution of the sample mean.

The analytical formula gives: **95% CI = mean ± 1.96 × (σ/√n)**

For the outdoor temperature dataset (n = 8,760, mean = 68.74°F, σ = 16.92°F):

Will the bootstrap CI and the analytical CI:

- **A)** Agree almost perfectly — close to the same interval
- **B)** Disagree noticeably — one method is more accurate

✅ Answer: A — They Agree Almost Perfectly

This validation is the whole point

Analytical CI: $68.74 \pm 1.96 \times \frac{16.92}{\sqrt{8760}} = (68.39, 69.09)$°F

Bootstrap CI (1,000 resamples): ≈ **(68.39, 69.09)**°F — within rounding error

They match because the Central Limit Theorem guarantees the sample mean is approximately Normal for large n — exactly what the analytical formula assumes.

> 💡 **Why validate at all?** Bootstrap is useful precisely because it works where no analytical formula exists (medians, percentiles, regression slopes, correlation coefficients). Validating it against a known formula proves the machinery works — then you can trust it when applied to harder statistics.

Bootstrap CI for Complex Statistics

No closed-form formula exists for the sampling distribution of a percentile or an R^2. Bootstrap handles both.

Example: CI for P95 of Cooling_kWh

The P95 of hourly cooling is a design criterion for chiller sizing (ASHRAE 90.1). Bootstrap gives you uncertainty bounds around that design value:

- Observed P95: the 95th percentile of all 8,760 hourly readings
- Bootstrap 95% CI: the 2.5th–97.5th percentile of 1,000 bootstrap P95 estimates

Bootstrap applies equally to the median, IQR, correlation coefficient, R^2, any regression coefficient, or any custom statistic — no formula derivation needed.

Pairs Bootstrap for Regression Slopes

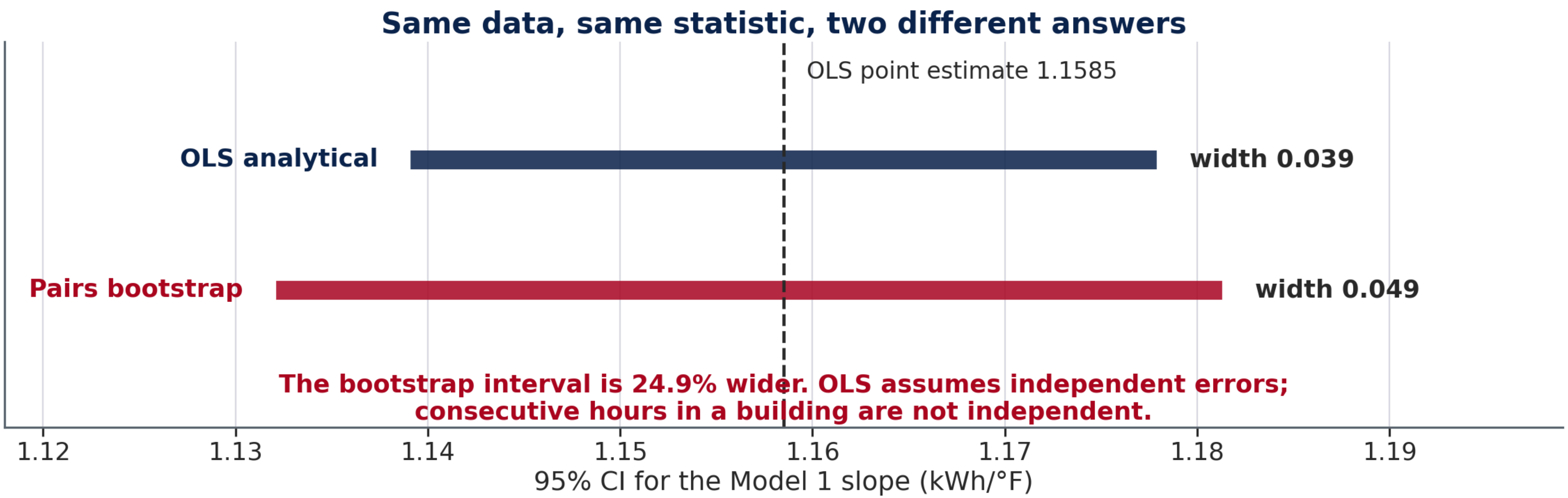
Resample whole **rows** — each row's (Temp, BH, kWh) stays together. Resampling x and y independently would destroy the correlation the regression exists to measure. Draw n rows with replacement, refit OLS, record the slope, repeat 1,000 times.

Both intervals are narrow at n = 8,760. But they do **not** agree:

95% CI for the Model 1 slope	Range (kWh/°F)	Width
OLS analytical	1.139 – 1.178	0.039
Pairs bootstrap	**1.132 – 1.181**	**0.049**

The bootstrap interval is **24.9 % wider**. OLS assumes independent errors; consecutive hours in a building are not. You compute that ratio yourself in Tutorial 14.

The Two Intervals, Side by Side



Both are 95% CIs for the same slope from the same 8,760 rows. The bootstrap is wider because it does not assume something about the data that is false.

Permutation Test — Assumption-Free Hypothesis Testing

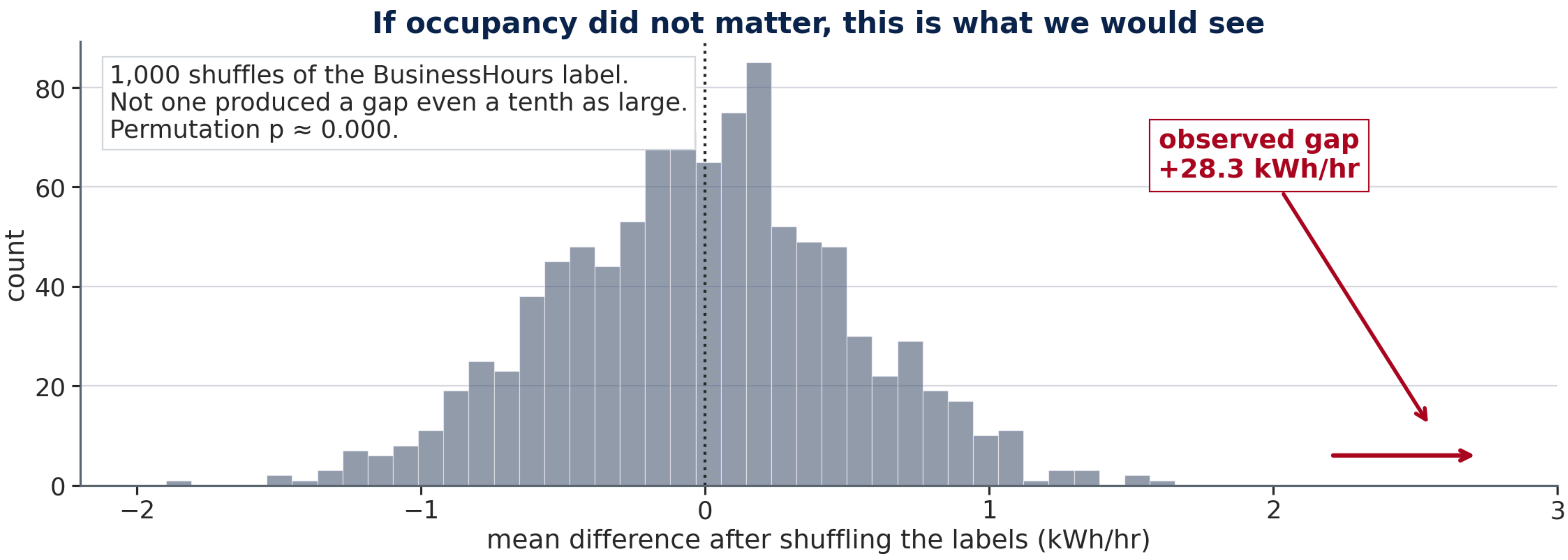
A permutation test answers: "if the group labels had no effect, how often would we see a difference as large as observed?"

For the BH effect on cooling (Week 11 revisited):

1. Observed: BH=1 mean – BH=0 mean ≈ +28.3 kWh/hr
2. Pool all 8,760 observations; randomly shuffle BH labels
3. Compute the shuffled mean difference
4. Repeat 1,000 times → how many shuffled differences ≥ 28.3?

For this dataset: **none** — the observed difference is far outside the permutation distribution. The permutation p-value is 0.000, matching the Week 11 t-test.

The Permutation Null



Shuffle the occupancy labels a thousand times and the mean gap never leaves ±2 kWh/hr. Ours was +28.3. The null distribution is built from the data, not assumed.

Quick Check — Is the Permutation Test Valid Here?

60 seconds · pick A or B

The Week 11 t-test assumed independence. We noted hourly building data is autocorrelated.

The **permutation test** works by randomly shuffling the BusinessHours labels 1,000 times under H_0 (no occupancy effect) and computing the mean difference each time.

Is the permutation test valid here, given the autocorrelation?

- **A)** Invalid — it still assumes independent observations
- **B)** Valid — it only assumes exchangeability under H_0, not independence

✅ Answer: B — Valid, Just Requires Exchangeability

Autocorrelation does not invalidate the permutation test

The permutation test's only assumption: **under H_0, the labels (BusinessHours = Yes/No) are exchangeable** — any assignment is equally likely if occupancy truly has no effect.

Autocorrelation means consecutive cooling values are correlated — but this is equally true for any label assignment under H_0. The test remains valid.

Method	Independence needed?	Result for BH test
t-test	Yes (approximate for large n)	$p \approx 0.000$
Permutation test	**No — only exchangeability**	**$p \approx 0.000$**

When Bootstrap and Analytical CIs Diverge

For this dataset (large n, approximately Normal residuals), bootstrap and analytical CIs agree closely. In other settings they can differ:

Situation	Effect on CI
Small n (< 30)	Bootstrap is unreliable; t-interval is better calibrated
Skewed distribution	Bootstrap CI is asymmetric; t-interval is symmetric
Heteroscedastic residuals	Bootstrap CI is wider; OLS SE understates uncertainty
Statistic with no formula	Only bootstrap applies

The key takeaway: for large, well-behaved datasets, both methods are valid. Bootstrap's advantage is generality — it works for any statistic regardless of whether a formula exists.

Limitations of Bootstrap

Bootstrap is powerful but not assumption-free:

- **Requires representative data** — if the sample is biased, the bootstrap CI captures the wrong population quantity
- **Cannot extrapolate** — bootstrap CI for P99 based on one year of data is unreliable if the population tail is heavier than observed
- **Autocorrelated data** — standard bootstrap assumes independent rows; hourly building energy data has autocorrelation (an unusually hot afternoon creates correlated cooling loads)
- **Computationally intensive** — 1,000 OLS fits is fast for $n = 8{,}760$; for $n = 1{,}000{,}000$ or complex models, runtime matters

For time-series applications (building energy, hydrology), a block bootstrap resamples contiguous segments to preserve autocorrelation structure.

Bridge to Machine Learning

Bootstrap concept	ML equivalent	Connection
Resample with replacement	Bagging (Bootstrap Aggregating)	Core of random forest training
Bootstrap CI for R^2	Model stability evaluation	How much does fit vary across folds?
Pairs bootstrap for slope	Cross-validation	Evaluate estimate stability on held-out data
Permutation test	Permutation feature importance	Shuffle one feature; measure accuracy drop

Random forest is bootstrap applied directly: each tree is trained on a bootstrap resample of the data. The ensemble prediction is the average across 100–1,000 trees, reducing variance compared to a single fit — exactly the same principle as averaging bootstrap resamples to estimate a statistic.

Looking Ahead — Week 15: Capstone Project

Every tool from Weeks 9–14, applied to **a dataset your team chooses**:

- **EDA** (Weeks 2–4): describe the distribution, the groups, the relationship
- **Regression** (Weeks 9–10): Model 1 (X alone) and Model 2 (X + your group variable)
- **Hypothesis test** (Week 11): is the group difference statistically real?
- **Bootstrap** (Week 14): a CI on your regression slope, no formula required
- **Monte Carlo** (Week 13): the range your outcome could take under uncertainty

Your team picks one numeric predictor **X**, one numeric outcome **Y**, and one **grouping variable** that splits the data in two — from any real civil or architectural engineering source. Five ready-made building-energy datasets remain available if you would rather not go looking.

Week 16: a **4-minute client briefing** plus 1 minute of questions, then a cross-team debrief. No Python tutorial in Week 15 — the lab *is* the capstone.